estimators based on Kalman filter theory can be designed (as seen in Section 10.4), these approaches are somewhat ad-hoc in that they attempt to modify an already existing approach. The H$_\infty$ filter was specifically designed for robustness.

In Section 11.1 we derive a different form of the Kalman filter and discuss the limitations of the Kalman filter. Section 11.2 discusses constrained optimization using Lagrange multipliers, which we will need later for our derivation of the H$_\infty$ filter. In Section 11.3 we use a game theory approach to derive the discrete-time H$_\infty$ filter, which minimizes the worst-case estimation error. This is in contrast to the Kalman filter's minimization of the expected value of the variance of the estimation error. Furthermore, the H$_\infty$ filter does not make any assumptions about the statistics of the process and measurement noise (although this information can be used in the H$_\infty$ filter if it is available). Section 11.4 presents the continuous-time H$_\infty$ filter, and Section 11.5 discusses an alternative method for deriving the H$_\infty$ filter using a transfer function approach.

11.1 INTRODUCTION

In this section we will first derive an alternate form for the Kalman filter. We do this to facilitate comparisons that we will make later in this chapter between the Kalman and H$_\infty$ filters. After we derive an alternate Kalman filter form, we will briefly discuss the limitations of the Kalman filter.

11.1.1 An alternate form for the Kalman filter

Recall that the Kalman filter estimates the state of a linear dynamic system defined by the equations

$$\begin{aligned} x_{k+1} &= F_k x_k + w_k \\ y_k &= H_k x + v_k \end{aligned} \tag{11.1}$$

where $\{w_k\}$ and $\{v_k\}$ are stochastic processes with covariances Q_k and R_k, respectively. As derived in Section 5.1, the Kalman filter equations are given as follows:

$$\begin{aligned} \hat{x}_{k+1}^- &= F_k \hat{x}_k^- + F_k K_k (y_k - H_k \hat{x}_k^-) \\ K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\ P_k^- &= F_{k-1} P_{k-1}^+ F_{k-1}^T + Q_{k-1} \\ P_k^+ &= (I - K_k H_k) P_k^- \end{aligned} \tag{11.2}$$

Using the matrix inversion lemma from Section 1.1.2 we see that

$$\begin{aligned} (H_k P_k^- H_k^T + R_k)^{-1} &= R_k^{-1} - R_k^{-1} H_k (\mathcal{I}_k^- + H_k^T R_k^{-1} H_k)^{-1} H_k^T R_k^{-1} \\ &= R_k^{-1} - R_k^{-1} H_k (I + P_k^- H_k^T R_k^{-1} H_k)^{-1} P_k^- H_k^T R_k^{-1} \end{aligned} \tag{11.3}$$

where $\mathcal{I}_k$ is the information matrix (i.e., the inverse of the covariance matrix P_k). The Kalman gain can therefore be written as follows:

$$\begin{aligned}
K_k &= P_k^- H_k^T (H_k P_k^- H_k^T + R_k)^{-1} \\
&= P_k^- H_k^T R_k^{-1} - P_k^- H_k^T R_k^{-1} H_k (I + P_k^- H_k^T R_k^{-1} H_k)^{-1} P_k^- H_k^T R_k^{-1} \\
&= \left[I - P_k^- H_k^T R_k^{-1} H_k (I + P_k^- H_k^T R_k^{-1} H_k)^{-1}\right] P_k^- H_k^T R_k^{-1} \\
&= \left[(I + P_k^- H_k^T R_k^{-1} H_k) - P_k^- H_k^T R_k^{-1} H_k\right] (I + P_k^- H_k^T R_k^{-1} H_k)^{-1} P_k^- H_k^T R_k^{-1} \\
&= (I + P_k^- H_k^T R_k^{-1} H_k)^{-1} P_k^- H_k^T R_k^{-1} \qquad (11.4)
\end{aligned}$$

Substituting this into the expression for P_{k+1}^- in Equation (11.2) we get

$$\begin{aligned}
P_{k+1}^- &= F_k P_k^+ F_k^T + Q_k \\
&= F_k (I - K_k H_k) P_k^- F_k^T + Q_k \\
&= F_k P_k^- F_k^T - F_k K_k H_k P_k^- F_k^T + Q_k \\
&= F_k P_k^- F_k^T - F_k (I + P_k^- H_k^T R_k^{-1} H_k)^{-1} P_k^- H_k^T R_k^{-1} H_k P_k^- F_k^T + Q_k \\
&= F_k P_k^- F_k^T - F_k (\mathcal{I}_k^- + H_k^T R_k^{-1} H_k)^{-1} H_k^T R_k^{-1} H_k P_k^- F_k^T + Q_k \qquad (11.5)
\end{aligned}$$

Apply the matrix inversion lemma again to the inverse on the right side of the above equation to obtain

$$\begin{aligned}
P_{k+1}^- &= F_k P_k^- F_k^T - \\
&\quad F_k \left[P_k^- - P_k^- H_k^T (R_k + H_k P_k^- H_k^T)^{-1} H_k P_k^-\right] H_k^T R_k^{-1} H_k P_k^- F_k^T + Q_k \\
&= F_k P_k^- \left[I - H_k^T R_k^{-1} H_k P_k^- + \right. \\
&\quad \left. H_k^T (R_k + H_k P_k^- H_k^T)^{-1} H_k P_k^- H_k^T R_k^{-1} H_k P_k\right] F_k^T + Q_k \\
&= F_k P_k^- T_k F_k^T + Q_k \qquad (11.6)
\end{aligned}$$

where T_k is defined by the above equation. Apply the matrix inversion lemma to the inverse that is in T_k to obtain

$$\begin{aligned}
T_k &= I - H_k^T R_k^{-1} H_k P_k^- + \\
&\quad H_k^T \left[R_k^{-1} - R_k^{-1} H_k (\mathcal{I}_k^- + H_k^T R_k^{-1} H_k)^{-1} H_k^T R_k^{-1}\right] H_k P_k^- H_k^T R_k^{-1} H_k P_k \\
&= I - H_k^T R_k^{-1} H_k P_k^- + (H_k^T R_k^{-1} H_k P_k^-)^2 - \\
&\quad H_k^T R_k^{-1} H_k (\mathcal{I}_k^- + H_k^T R_k^{-1} H_k)^{-1} (H_k^T R_k^{-1} H_k P_k^-)^2 \\
&= I - H_k^T R_k^{-1} H_k P_k^- + (H_k^T R_k^{-1} H_k P_k^-)^2 - \\
&\quad H_k^T R_k^{-1} H_k P_k^- (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} (H_k^T R_k^{-1} H_k P_k^-)^2 \\
&= I - H_k^T R_k^{-1} H_k P_k^- + (H_k^T R_k^{-1} H_k P_k^-)^2 - \\
&\quad (H_k^T R_k^{-1} H_k P_k^-)^3 (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} \\
&= \left[(I + H_k^T R_k^{-1} H_k P_k^-) - H_k^T R_k^{-1} H_k P_k^- (I + H_k^T R_k^{-1} H_k P_k^-) + \right. \\
&\quad \left. (H_k^T R_k^{-1} H_k P_k^-)^2 (I + H_k^T R_k^{-1} H_k P_k^-) - (H_k^T R_k^{-1} H_k P_k^-)^3\right] \times \\
&\quad (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} \\
&= (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} \qquad (11.7)
\end{aligned}$$

Substituting this expression for T_k into Equation (11.6) gives

$$P_{k+1}^- = F_k P_k^- (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} F_k^T + Q_k \qquad (11.8)$$

From Equation (11.4) the Kalman gain can be written as

$$K_k = (I + P_k^- H_k^T R_k^{-1} H_k)^{-1} P_k^- H_k^T R_k^{-1} \tag{11.9}$$

We can premultiply outside the parentheses by P_k^-, and postmultiply each term inside the parenthesis by P_k^-, to obtain

$$K_k = P_k^- (P_k^- + P_k^- H_k^T R_k^{-1} H_k P_k^-)^{-1} P_k^- H_k^T R_k^{-1} \tag{11.10}$$

We can postmultiply outside the parentheses by the inverse of P_k^-, and premultiply each term inside the parentheses by the inverse of P_k^-, to obtain

$$K_k = P_k^- (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} H_k^T R_k^{-1} \tag{11.11}$$

Combining this expression for K_k with Equations (11.2) and (11.8) we can summarize the Kalman filter as follows:

$$\begin{aligned}
\hat{x}_{k+1}^- &= F_k \hat{x}_k^- + F_k K_k (y_k - H_k \hat{x}_k^-) \\
K_k &= P_k^- (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} H_k^T R_k^{-1} \\
P_{k+1}^- &= F_k P_k^- (I + H_k^T R_k^{-1} H_k P_k^-)^{-1} F_k^T + Q_k
\end{aligned} \tag{11.12}$$

11.1.2 Kalman filter limitations

The Kalman filter works well, but only under certain conditions.

- First, we need to know the mean and correlation of the noise w_k and v_k at each time instant.
- Second, we need to know the covariances Q_k and R_k of the noise processes. The Kalman filter uses Q_k and R_k as design parameters, so if we do not know Q_k and R_k then it may be difficult to successfully use a Kalman filter.
- Third, the attractiveness of the Kalman filter lies in the fact that it is the one estimator that results in the smallest possible standard deviation of the estimation error. That is, the Kalman filter is the minimum variance estimator if the noise is Gaussian, and it is the linear minimum variance estimator if the noise is not Gaussian. If we desire to minimize a different cost function (such as the worst-case estimation error) then the Kalman filter may not accomplish our objectives.
- Finally, we need to know the system model matrices F_k and H_k.

So what do we do if one of the Kalman filter assumptions is not satisfied? What should we do if we do not have any information about the noise statistics? What should we do if we want to minimize the worst-case estimation error rather than the covariance of the estimation error?

Perhaps we could just use the Kalman filter anyway, even though its assumptions are not satisfied, and hope for the best. That is a common solution to our Kalman filter quandary and it works reasonably well in many cases. However, there is yet another option that we will explore in this chapter: the H$_\infty$ filter, also called the minimax filter. The H$_\infty$ filter does not make any assumptions about the noise, and it minimizes the worst-case estimation error (hence the term minimax).

11.2 CONSTRAINED OPTIMIZATION

In this section we show how constrained optimization can be performed through the use of Lagrange multipliers. This background is required for the solution of the H_∞ filtering problem that is presented in Section 11.3. In Section 11.2.1 we will investigate static problems (i.e., problems in which the independent variables are constant). In Section 11.2.2 we will take a brief segue to look at problems with inequality constraints. In Section 11.2.3 we will extend our constrained optimization method to dynamic problems (i.e., problems in which the independent variables change with time).

11.2.1 Static constrained optimization

Suppose we want to minimize some scalar function $J(x, w)$ with respect to x and w. x is an n-dimensional vector, and w is an m-dimensional vector. w is the independent variable and x is the dependent variable; that is, x is somehow determined by w. Suppose our vector-valued constraint is given as $f(x, w) = 0$. Further assume that the dimension of $f(x, w)$ is the same as the dimension of x. This problem can be written as

$$\min_{x,w} J(x, w) \text{ such that } f(x, w) = 0 \tag{11.13}$$

Suppose that the constrained minimum of $J(x, w)$ occurs at $x = x^*$ and $w = w^*$. We call this the stationary point of $J(x, w)$. Now suppose that we choose values of x and w such that x is close to x^*, w is close to w^*, and $f(x, w) = 0$. Expanding $J(x, w)$ and $f(x, w)$ in a Taylor series around x^* and w^* gives

$$\begin{aligned} J(x, w) &= J(x^*, w^*) + \left.\frac{\partial J}{\partial x}\right|_{x^*,w^*} \Delta x + \left.\frac{\partial J}{\partial w}\right|_{x^*,w^*} \Delta w \\ f(x, w) &= f(x^*, w^*) + \left.\frac{\partial f}{\partial x}\right|_{x^*,w^*} \Delta x + \left.\frac{\partial f}{\partial w}\right|_{x^*,w^*} \Delta w \end{aligned} \tag{11.14}$$

where higher-order terms have been neglected (with the assumption that x is close to x^*, and w is close to w^*), $\Delta x = x - x^*$, and $\Delta w = w - w^*$. These equations can be written as

$$\begin{aligned} \Delta J(x, w) &= J(x, w) - J(x^*, w^*) \\ &= \left.\frac{\partial J}{\partial x}\right|_{x^*,w^*} \Delta x + \left.\frac{\partial J}{\partial w}\right|_{x^*,w^*} \Delta w \\ \Delta f(x, w) &= f(x, w) - f(x^*, w^*) \\ &= \left.\frac{\partial f}{\partial x}\right|_{x^*,w^*} \Delta x + \left.\frac{\partial f}{\partial w}\right|_{x^*,w^*} \Delta w \end{aligned} \tag{11.15}$$

Now note that for values of x and w that are close to x^* and w^*, we have $\Delta J(x, w) = 0$. This is because the partial derivatives on the right side of the $\Delta J(x, w)$ equation are zero at the stationary point of $J(x, w)$. We also see that $\Delta f(x, w) = 0$ at the stationary point of $J(x, w)$. This is because $f(x^*, w^*) = 0$ at the constrained stationary point of $J(x, w)$, and we chose x and w such that $f(x, w) = 0$ also. The above equations can therefore be written as

$$\begin{aligned}
\left.\frac{\partial J}{\partial x}\right|_{x^*,w^*} \Delta x + \left.\frac{\partial J}{\partial w}\right|_{x^*,w^*} \Delta w &= 0 \\
\left.\frac{\partial f}{\partial x}\right|_{x^*,w^*} \Delta x + \left.\frac{\partial f}{\partial w}\right|_{x^*,w^*} \Delta w &= 0
\end{aligned} \tag{11.16}$$

These equations are true for arbitrary x and w that are close to x^* and w^* and that satisfy the constraint $f(x, w) = 0$. Equation (11.16) can be solved for Δx as

$$\Delta x = -\left(\left.\frac{\partial f}{\partial x}\right|_{x^*,w^*} \right)^{-1} \left.\frac{\partial f}{\partial w}\right|_{x^*,w^*} \Delta w \tag{11.17}$$

This can be substituted into Equation (11.16) to obtain

$$\left.\frac{\partial J}{\partial w}\right|_{x^*,w^*} - \left.\frac{\partial J}{\partial x}\right|_{x^*,w^*} \left(\left.\frac{\partial f}{\partial x}\right|_{x^*,w^*} \right)^{-1} \left.\frac{\partial f}{\partial w}\right|_{x^*,w^*} = 0 \tag{11.18}$$

This equation, combined with the constraint $f(x, w) = 0$, gives us $(m+n)$ equations that can be solved for the vectors w and x to find the constrained stationary point of $J(x, w)$.

Now consider the augmented cost function

$$J_a = J + \lambda^T f \tag{11.19}$$

where λ is an n-element unknown constant vector called a Lagrange multiplier. Note that

$$\begin{aligned}
\frac{\partial J_a}{\partial x} &= \frac{\partial J}{\partial x} + \lambda^T \frac{\partial f}{\partial x} \\
\frac{\partial J_a}{\partial w} &= \frac{\partial J}{\partial w} + \lambda^T \frac{\partial f}{\partial w} \\
\frac{\partial J_a}{\partial \lambda} &= f
\end{aligned} \tag{11.20}$$

If we set all three of these equations equal to zero then we have

$$\begin{aligned}
\lambda^T &= \frac{-\partial J}{\partial x} \left(\frac{\partial f}{\partial x} \right)^{-1} \\
\frac{\partial J}{\partial w} - \frac{\partial J}{\partial x} \left(\frac{\partial f}{\partial x} \right)^{-1} \frac{\partial f}{\partial w} &= 0 \\
f &= 0
\end{aligned} \tag{11.21}$$

The first equation gives us the value of the Lagrange multiplier, the second equation is identical to Equation (11.18), and the third equation forces the constraint to be satisfied. We therefore see that we can solve the original constrained problem by creating an augmented cost function J_a, taking the partial derivatives with respect to x, w, and λ, setting them equal to zero, and solving for x, w, and λ. The partial derivative equations give us $(2n + m)$ equations to solve for the n-element vector x, the m-element vector w, and the n-element vector λ. We have increased the dimension of the original problem by introducing a Lagrange multiplier, but we have transformed the constrained optimization problem into an unconstrained optimization problem, which can simplify the problem considerably.

■ **EXAMPLE 11.1**

Suppose we need to find the minimum of $J(x, u) = x^2/2 + xu + u^2 + u$ with respect to x and u such that $f(x, u) = x - 3 = 0$. This simple example can be solved by simply realizing that $x = 3$ in order to satisfy the constraint. Substituting $x = 3$ into $J(x, u)$ gives $J(x, u) = 9/2 + 4u + u^2$. Setting the derivative with respect to u equal to zero and solving for u gives $u = -2$.

We can also solve this problem using the Lagrange multiplier method. We create an augmented cost function as

$$\begin{aligned} J_a &= J + \lambda^T f \\ &= x^2/2 + xu + u^2 + u + \lambda(x - 3) \end{aligned} \tag{11.22}$$

The Lagrange multiplier λ has the same dimension as x (scalar in this example). The three necessary conditions for a constrained stationary point of J are obtained by setting the partial derivations of Equation (11.20) equal to 0.

$$\begin{aligned} \frac{\partial J_a}{\partial x} &= x + u + \lambda = 0 \\ \frac{\partial J_a}{\partial u} &= x + 2u + 1 = 0 \\ \frac{\partial J_a}{\partial \lambda} &= x - 3 = 0 \end{aligned} \tag{11.23}$$

Solving these three equations for x, u, and λ gives $x = 3$, $u = -2$, and $\lambda = -1$. In this example the Lagrange multiplier method seems to require more effort than simply solving the problem directly. However, in more complicated constrained optimization problems the Lagrange multiplier method is essential for finding a solution.

$\nabla\nabla\nabla$

11.2.2 Inequality constraints

Suppose that we want to minimize a scalar function that is subject to an inequality constraint:

$$\min J(x) \text{ such that } f(x) \leq 0 \tag{11.24}$$

This can be reduced to two minimization problems, neither of which contain inequality constraints. The first minimization problem is unconstrained, and the second minimization problem has an equality constraint:

1. $\min J(x)$
2. $\min J(x)$ such that $f(x) = 0$

In other words, the optimal value of x is either not on the constraint boundary [i.e., $f(x) < 0$], or it is on the constraint boundary [i.e., $f(x) = 0$]. If it is not on the constraint boundary then $f(x) < 0$ and the optimal value of x is obtained by solving the problem without the constraint. If it is on the constraint boundary then $f(x) = 0$ at the constrained minimum, and the optimal value of x is obtained by solving the problem with the equality constraint $f(x) = 0$.

The procedure for solving Equation (11.24) involves solving the unconstrained problem first. Then we check to see if the unconstrained minimum satisfies the constraint. If the unconstrained minimum satisfies the constraint, then the unconstrained minimum solves the inequality-constrained minimization problem and we are done. However, if the unconstrained minimum does not satisfy the constraint, then the minimization problem with the inequality constraint is equivalent to the minimization problem with the equality constraint. So we solve the problem with the equality constraint $f(x) = 0$ to obtain the final solution. This is illustrated for the scalar case in Figure 11.1.

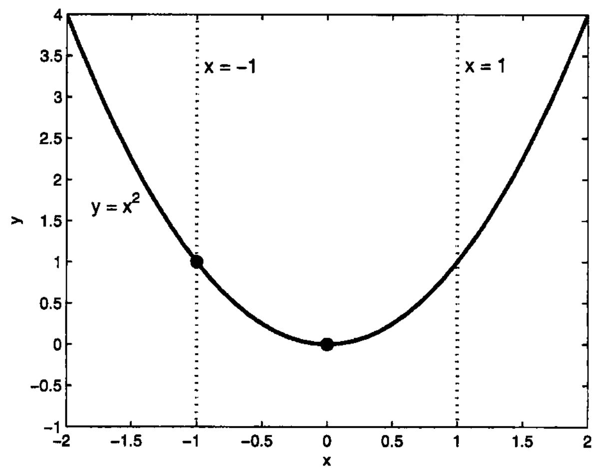

Figure 11.1 This illustrates the constrained minimization of x^2. If the constraint is $x - 1 \leq 0$, then the constrained minimum is equal to the unconstrained minimum and occurs at $x = 0$. If the constraint is $x + 1 \leq 0$, then the constrained minimum can be solved by enforcing the equality constraint $x + 1 = 0$ and occurs at $x = -1$.

When we extend this idea to more than one dimension, we obtain the following procedure, which is called the active-set method for optimization with inequality constraints [Fle81, Gil81].

1. The problem is to minimize $J(x)$ such that $f(x) \leq 0$, where $f(x)$ is an m-element constraint function and the inequality is taken one element at a time.

2. First solve the unconstrained minimization problem. If the unconstrained solution satisfies the constraint $f(x) \leq 0$ then the problem is solved. If not, continue to the next step.

3. For all possible combinations of constraints, solve the problem using those constraints as equality constraints. If the solution satisfies the remaining (unused) constraints, then the solution is feasible. Note that this step requires the solution of $(2^m - 1)$ constrained optimization problems.

4. Out of all the feasible solutions that were obtained in the previous step, the one with the smallest $J(x)$ is the solution to the constrained minimization problem.

Note that there are also other methods for solving optimization problems with inequality constraints, including primal–dual interior-point methods [Wri97].

11.2.3 Dynamic constrained optimization

In this section we extend the Lagrange multiplier method of constrained optimization to the optimization of dynamic systems. Suppose that we have a dynamic system given as

$$x_{k+1} = F_k x_k + w_k \qquad (k = 0, \cdots, N-1) \tag{11.25}$$

where x_k is an n-dimensional state vector. We want to minimize the scalar function

$$J = \psi(x_0) + \sum_{k=0}^{N-1} \mathcal{L}_k \tag{11.26}$$

where $\psi(x_0)$ is a known function of x_0, and $\mathcal{L}_k$ is a known function of x_k and w_k. This is a constrained dynamic optimization problem similar to the type that arises in optimal control [Lew86a, Ste94]. It is slightly different than typical optimal control problems because $\psi(x_k)$ in the above equation is evaluated at the initial time ($k = 0$) instead of the final time ($k = N$), but the methods of optimal control can be used with only slight modifications to solve our problem. The constraints are given in Equation (11.25). From the previous section we know that we can solve this problem by introducing a Lagrange multiplier λ, creating an augmented cost function J_a, and then setting the partial derivatives of J_a with respect to x_k, w_k, and λ equal to zero. Since we have N constraints in Equation (11.25) (each of dimension n), we have to introduce N Lagrange multipliers $\lambda_1, \cdots, \lambda_N$ (each of dimension n). The augmented cost function is therefore written as

$$J_a = \psi(x_0) + \sum_{k=0}^{N-1} \left[\mathcal{L}_k + \lambda_{k+1}^T (F_k x_k + w_k - x_{k+1})\right] \tag{11.27}$$

This can be written as

$$\begin{aligned} J_a &= \psi(x_0) + \sum_{k=0}^{N-1} \left[\mathcal{L}_k + \lambda_{k+1}^T (F_k x_k + w_k)\right] - \sum_{k=0}^{N-1} \lambda_{k+1}^T x_{k+1} \\ &= \psi(x_0) + \sum_{k=0}^{N-1} \left[\mathcal{L}_k + \lambda_{k+1}^T (F_k x_k + w_k)\right] - \sum_{k=0}^{N} \lambda_k^T x_k + \lambda_0^T x_0 \end{aligned} \tag{11.28}$$

where λ_0 is now an additional term in the Lagrange multiplier sequence. It is not in the original augmented cost function, but we will see in Section 11.3 that its value will be determined when we solve the constrained optimization problem. Now we define the Hamiltonian $\mathcal{H}_k$ as

$$\mathcal{H}_k = \mathcal{L}_k + \lambda_{k+1}^T (F_k x_k + w_k) \tag{11.29}$$

With this notation we can write the augmented cost function as follows.

$$
\begin{aligned}
J_a &= \psi(x_0) + \sum_{k=0}^{N-1} \mathcal{H}_k - \sum_{k=0}^{N} \lambda_k^T x_k + \lambda_0^T x_0 \\
&= \psi(x_0) + \sum_{k=0}^{N-1} \mathcal{H}_k - \sum_{k=0}^{N-1} \lambda_k^T x_k - \lambda_N^T x_N + \lambda_0^T x_0 \\
&= \psi(x_0) + \sum_{k=0}^{N-1} \left(\mathcal{H}_k - \lambda_k^T x_k\right) - \lambda_N^T x_N + \lambda_0^T x_0 \qquad (11.30)
\end{aligned}
$$

The conditions that are required for a constrained stationary point are

$$
\begin{aligned}
\frac{\partial J_a}{\partial x_k} &= 0 \quad (k = 0, \cdots, N) \\
\frac{\partial J_a}{\partial w_k} &= 0 \quad (k = 0, \cdots, N-1) \\
\frac{\partial J_a}{\partial \lambda_k} &= 0 \quad (k = 0, \cdots, N) \qquad (11.31)
\end{aligned}
$$

These conditions can also be written as

$$
\begin{aligned}
\frac{\partial J_a}{\partial x_0} &= 0 \\
\frac{\partial J_a}{\partial x_N} &= 0 \\
\frac{\partial J_a}{\partial x_k} &= 0 \quad (k = 1, \cdots, N-1) \\
\frac{\partial J_a}{\partial w_k} &= 0 \quad (k = 0, \cdots, N-1) \\
\frac{\partial J_a}{\partial \lambda_k} &= 0 \quad (k = 0, \cdots, N) \qquad (11.32)
\end{aligned}
$$

The fifth condition ensures that the constraint $x_{k+1} = F_k x_k + w_k$ is satisfied. Based on the expression for J_a in Equation (11.30), the first four conditions above can be written as

$$
\begin{aligned}
\lambda_0^T + \frac{\partial \psi_0}{\partial x_0} &= 0 \\
-\lambda_N^T &= 0 \\
\lambda_k^T &= \frac{\partial \mathcal{H}_k}{\partial x_k} \quad (k = 1, \cdots, N-1) \\
\frac{\partial \mathcal{H}_k}{\partial w_k} &= 0 \quad (k = 0, \cdots, N-1) \qquad (11.33)
\end{aligned}
$$

This gives us the necessary conditions for a constrained stationary point of our dynamic optimization problem. These are the results that we will use to solve the H$_\infty$ estimation problem in the next section.